MTG Commander Recommender System

Group 1 – ManaCurve Analytics

Data Science Capstone Project
Final Report

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Analysis the basic metrics of variables

The cleaned Commander-legal dataset contains 30,465 cards and 11 variables. Each row represents a unique card after deduplication across printings. The variables include identifiers (name, uuid), numerical variables (manaValue, power, toughness), categorical variables (rarity), multivalued categorical variables (colors, colorIdentity, types), a boolean variable (isLegendary), and a text variable (text).

The numerical variables manaValue, power, and toughness are discrete quantitative variables, although stored as floats. Mana value applies to all cards, while power and toughness apply only to creatures. The categorical variable rarity is ordinal, as it reflects an inherent hierarchy (common < uncommon < rare < mythic < special). The multivalued variables colors, colorIdentity, and types are nominal categorical variables without inherent ordering. The isLegendary variable is a nominal boolean indicator identifying whether a card has the "Legendary" supertype.

Missing values are concentrated in power (44.85%) and toughness (44.69%), which is expected because non-creature cards do not have these attributes. The text field contains 1.15% missing values. All other variables are complete.

For manaValue (n = 30,465), the mean is 3.29 with a standard deviation of 1.76. The minimum value is 0 and the maximum is 16. The 25th percentile is 2, the median is 3, and the 75th percentile is 4, indicating that most cards fall within the 2–4 mana range. The distribution shows moderate positive skew (0.61), reflecting a concentration of low-cost cards with fewer high-cost outliers.

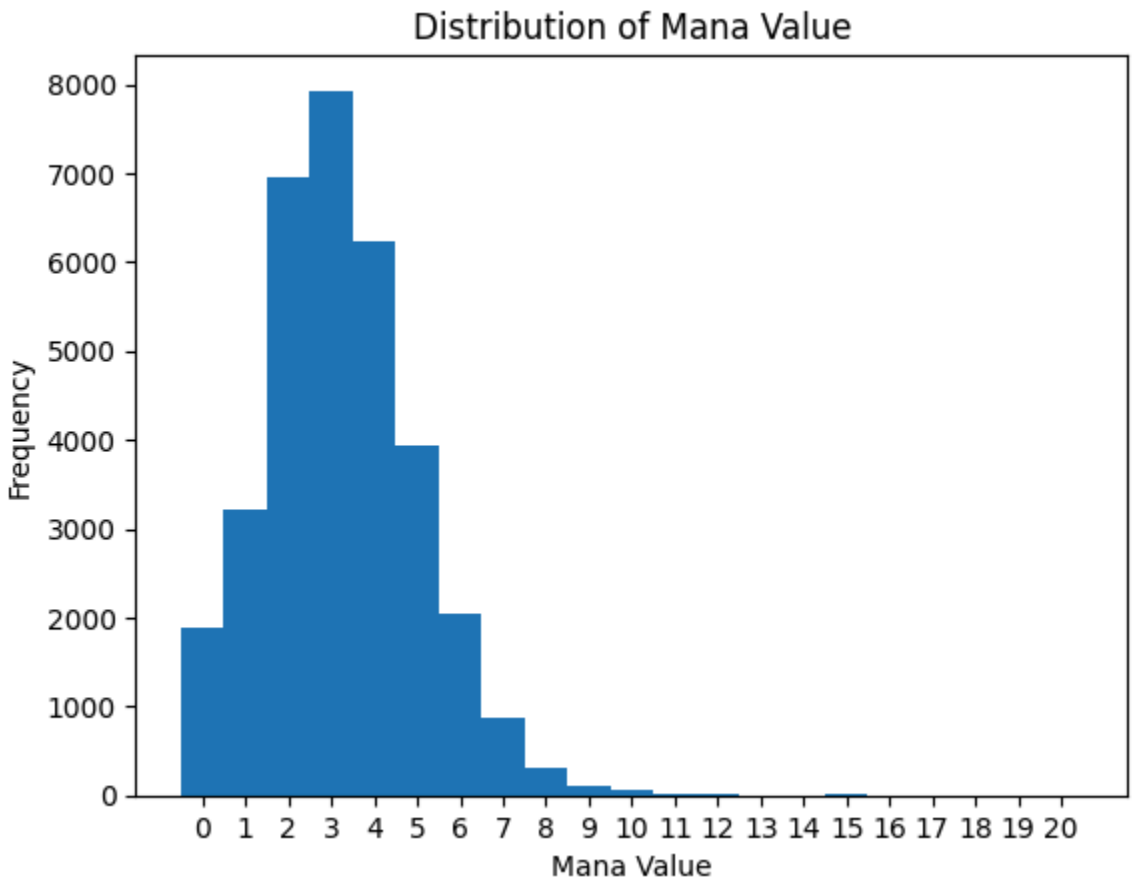
For creature cards, power (n = 16,800) has a mean of 2.67 and a standard deviation of 1.69, with values ranging from -1 to 20. The 25th percentile is 2, the median is 2, and the 75th percentile is 3. Toughness (n = 16,851) has a mean of 2.89 and a standard deviation of 1.70, with values ranging from -1 to 30. The 25th percentile is 2, the median is 3, and the 75th percentile is 4. The minimum value of -1 reflects special variable-stat cards that were coerced to numeric form.

In terms of categorical distributions, commons represent 32.89% of the dataset, rares 30.89%, uncommons 30.02%, mythics 6.16%, and special rarity cards 0.03%. Mono-colored cards dominate the dataset (22,648), followed by colorless cards (3,826) and two-color cards (3,278), with higher color combinations becoming increasingly rare. Creatures are the most common card type (16,816), followed by instants, enchantments, artifacts, and sorceries. Finally, 11.34% of cards are legendary, which is particularly relevant for Commander deck construction.

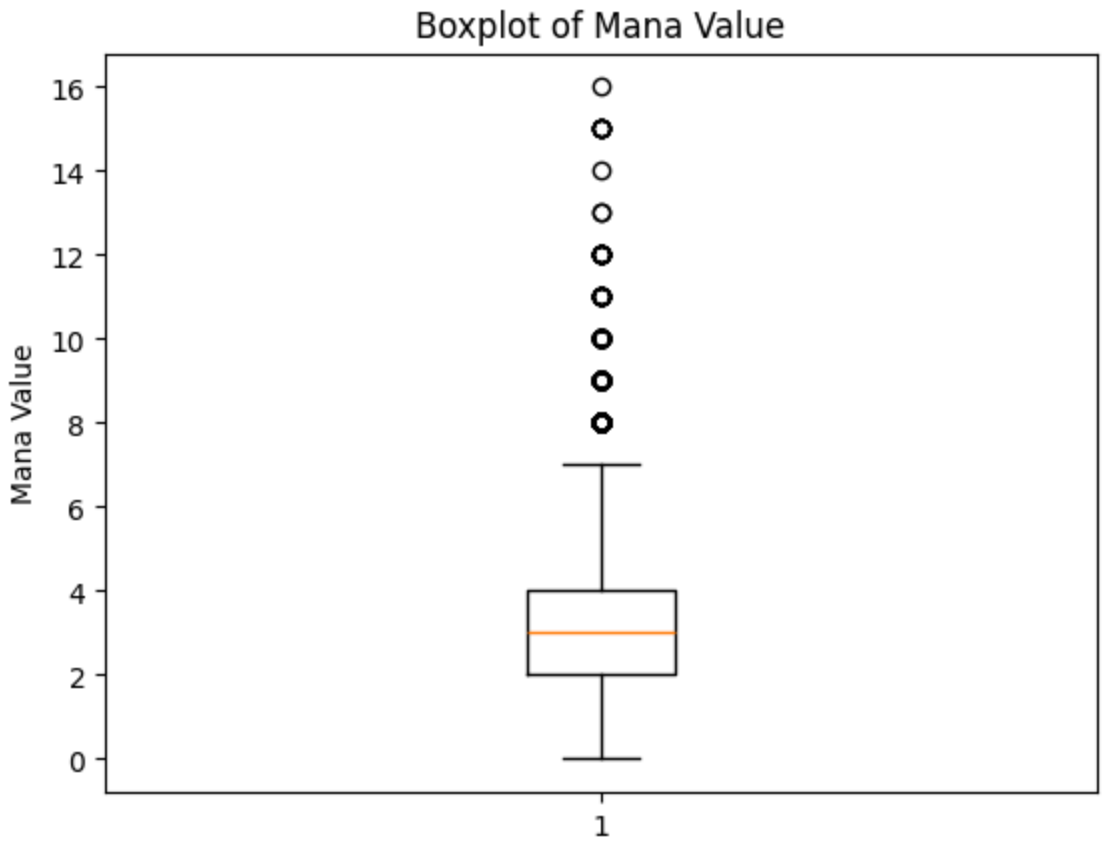
Overall, the dataset contains a structured mix of discrete numerical variables, ordinal and nominal categorical variables, multivalued categorical attributes, and text features. The numerical variables are concentrated within relatively narrow ranges, and several categorical variables exhibit natural imbalance consistent with game design.

Non-graphical and graphical univariate analysis

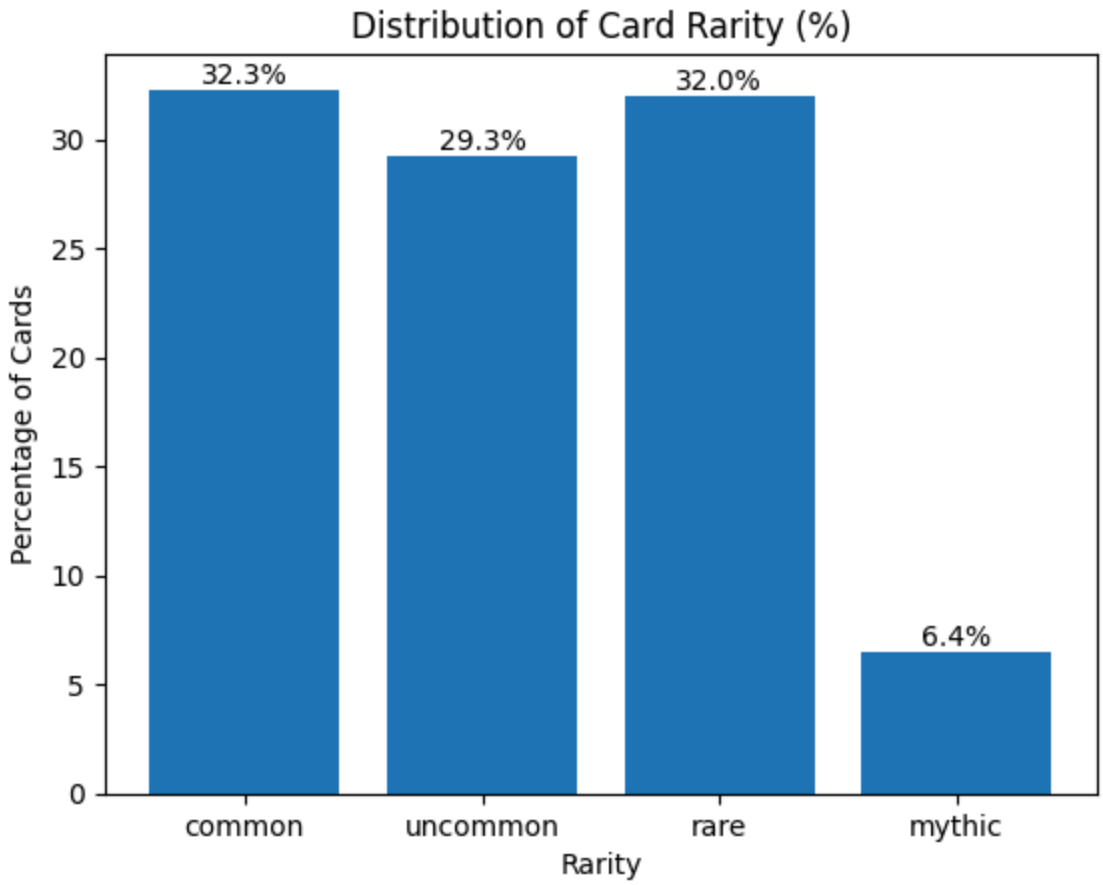
In the de-duplicated dataset, mana value is primarily concentrated between 1 and 5, with a clear peak around 2–4. The distribution is right-skewed with a long upper tail, indicating that high-cost cards exist but are substantially less common than low-cost cards. A small number of zero-cost entries and extreme outliers (e.g., very large mana values) are also present; these likely correspond to special card mechanics or encoding/placeholder values rather than typical gameplay costs.



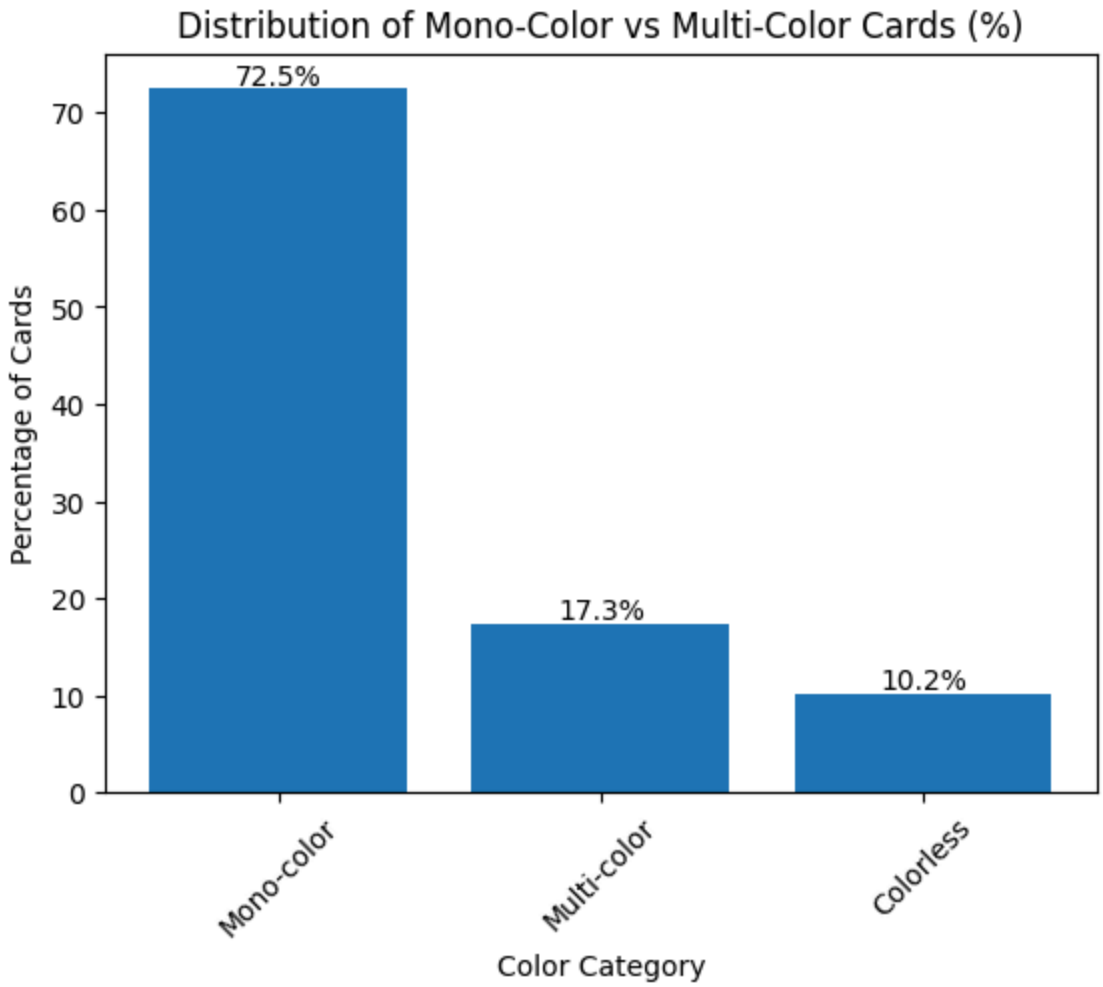
The box-and-whisker plot shows that mana values are tightly concentrated in the low-to-mid range: the lower quartile (Q1) is about 2, the median is 3, and the upper quartile (Q3) is about 4, giving an interquartile range (IQR) of roughly 2. The lower whisker reaches down to 0 while the upper whisker extends to about 7, but there is a long string of clear outliers above the upper whisker (spread from ~8 up to ~16). This shape, a short box, a longer upper whisker, and many high-value outliers — visually confirms a right-skewed distribution with most cards clustered at 2–4 mana and a small number of high-cost cards stretching the upper tail. The isolated zero-cost points and extreme high-cost values likely represent special card mechanics or encoding/placeholder values rather than typical gameplay costs; they should be inspected and either documented, recoded, or treated separately (e.g., flagged as special cases or winsorized) before downstream modeling.



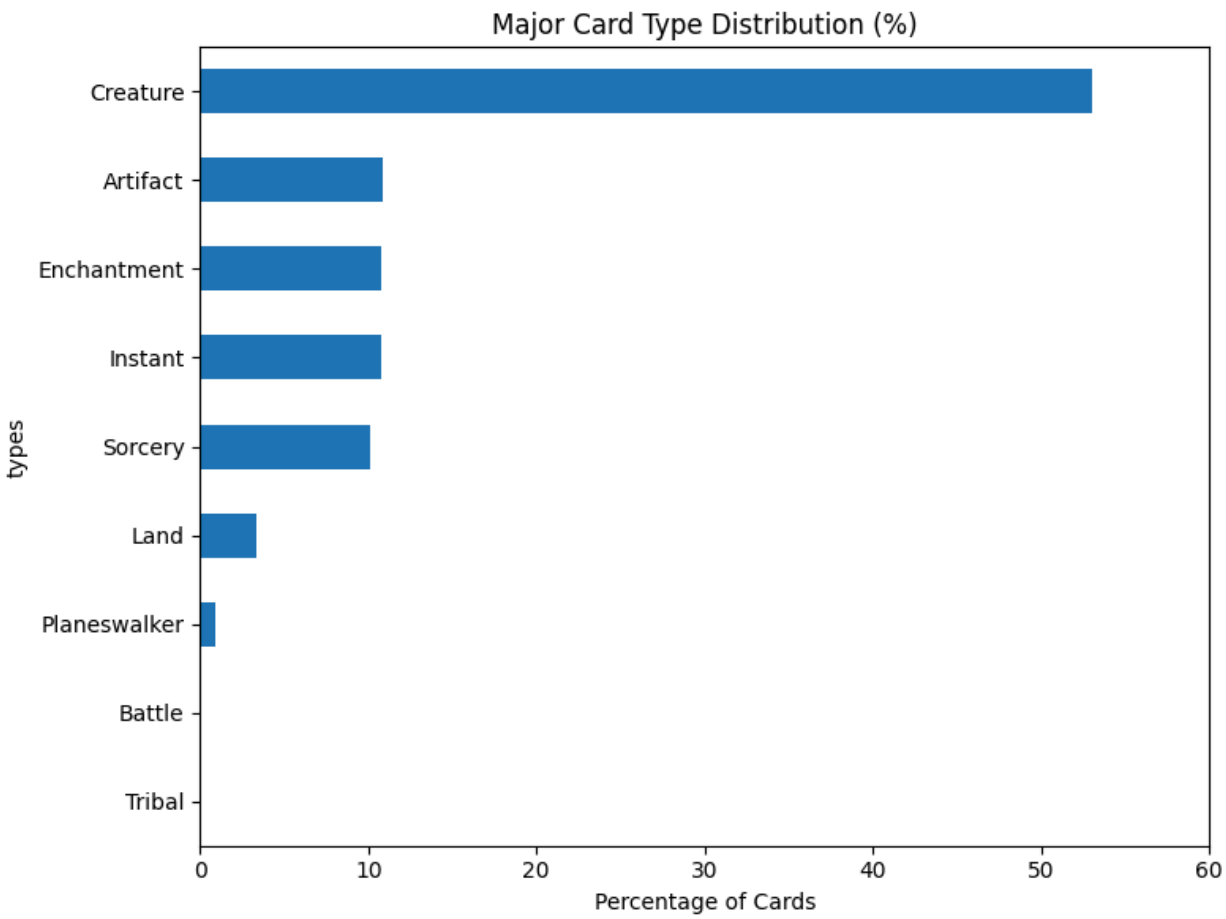
Uncommon, and Rare making up the vast majority of cards, while Mythic represents only a small fraction. This indicates that higher-rarity cards are intentionally scarce relative to standard cards. Overall, the distribution reflects a typical design pattern in which most cards are broadly accessible, while a limited subset is reserved for high-rarity categories.



The chart indicates that the card pool is overwhelmingly mono-color, accounting for approximately **72.5%** of cards, while multi-color cards make up about **17.3%** and colorless cards about **10.2%**. This distribution suggests the set strongly emphasizes single-color identity, with multi-color and colorless cards serving as less common but supporting elements within the overall design.



Creature cards make up approximately **53.6%** of the dataset. Artifact (**11%**), Enchantment (**10.9%**), Instant (**10.9%**), and Sorcery (**10.3%**) each represent around one-tenth of cards, meaning these five major types together account for the overwhelming majority of the pool. Land cards are much less common at about **3.4%**, while Planeswalker (**0.9%**), Battle (**0.1%**), and Tribal (**0.0%**) appear only rarely. This distribution indicates a strong emphasis on creature-based gameplay, with other card types serving supporting roles and several categories existing as niche mechanics.



Missing value analysis and outlier analysis

As previously noted, the majority of missing values are for the power and toughness variables at around 45% each. This is because power and toughness are only relevant for creature cards; non-creature cards do not have either attribute. Because the missing values for power and toughness are structurally missing for non-creature cards, no imputation or alternative approach is needed. NaN values will be left for non-creature cards.

The Text field contains 1.15% missing values. These cards represent cards with no rules rather than incomplete data. To help facilitate any natural language processing, missing text entries are replaced with empty strings rather than removed, allowing us to preserve the full dataset for analysis. All of the other variables are complete.

To perform analysis on outliers in the data, we use the IQR method. This approach is used because we are dealing with intentionally structured data from a game rather than a dataset with measurement error. The distributions are skewed, and an alternative approach like Z-scores assume that there is approximately a normal distribution. Using the IQR method, observations below Q1 - 1.5xIQR or above Q3 + 1.5xIQR are classified as extreme values.

After implementing the IQR method on Mana value, power, and toughness, there was a small percentage of observations flagged as extreme values. After inspecting these cases, they were confirmed to be legitimate high cost spells and creatures with high stats rather than data entry errors. These values are intentional for the game design and represent meaningful structural variation of the different cards and rarities in the Commander format. Therefore, no observations were removed following the outlier analysis.

Feature engineering and analysis

In order to support predictive modeling and the development of a content-based recommender system, several additional variables were engineered from the cleaned dataset. The objective of feature engineering was to convert multivalued categorical variables and text attributes into structured numeric representations suitable for statistical analysis and machine learning models. These transformations allow card attributes such as color, identity, card role, and mechanical complexity to be incorporated into similarity calculations and predictive algorithms.

Several variables in the dataset contain multiple values per observation. For example, the colors and types attributes are multivalued categorical variables because a single card can simultaneously belong to multiple categories (e.g Land Creature). A multi-hot encoding strategy was used to represent these attributes numerically.

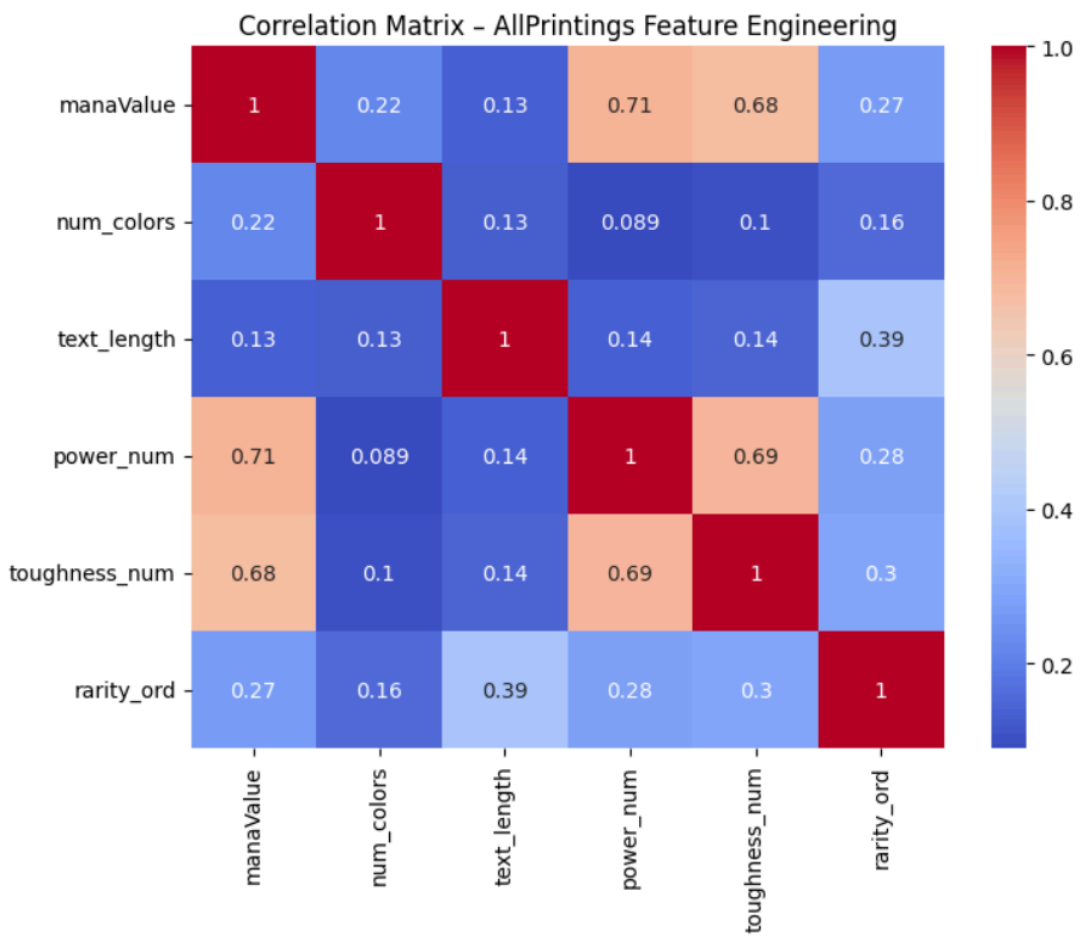
For color identity, five binary indicator variables were created (color_W, color_U, color_B, color_R, and color_G) corresponding to the five Magic: The Gathering colors. Each variable takes the value 1 if the card includes that color and 0 otherwise. This representation preserves the possibility of multi-color cards while allowing color information to be incorporated into machine learning algorithms. Similarly, binary indicator variables were generated for the major card types (Creature, Instant, Sorcery, Artifact, Enchantment, and Land). This representation allows for cards to belong to multiple categories simultaneously while allowing models to capture functional similarities between cards.

In addition to categorical encodings, several derived numerical variables were created to capture important card characteristics. The num_colors variable was derived by counting the number of colors associated with each card. This feature distinguishes mono-colored, multi-colored, and colorless cards and provides a measure of color complexity that is particularly relevant in Commander deck construction. The length of the rules text was used as a proxy for mechanical complexity and is stored in the text_length variable.

The categorical rarity variable was also transformed into an ordinal numerical representation. The categories common, uncommon, rare, and mythic were encoded as increasing integer values to reflect the natural hierarchy between rarity levels. This transformation allows rarity to be incorporated into correlation analysis and machine learning models while preserving ordinal structure. Finally, the combat statistics power and toughness were converted to numeric variables (power_num and toughness_num) to ensure consistent numeric representation.

Following feature engineering, correlation analysis was conducted among the numeric variables manaValue, num_colors, text_length, power_num, toughness_num, and rarity_ord. The analysis revealed several meaningful relationships within the data. Mana value exhibited strong positive correlation with

both power and toughness, reflecting the underlying design principle that higher-cost creatures tend to have stronger combat statistics. Power and toughness were also strongly correlated with each other, which is expected because creature strength generally scales proportionally in both offensive and defensive capabilities. A moderate positive correlation was observed between rarity and text_length, suggesting that higher rarity cards tend to have more complex rules text and mechanics. This relationship aligns with the design philosophy of the game, where more mechanically complex cards are often placed at higher rarity levels. Other variables such as num_colors, exhibited relatively weak correlations with the remaining features, indicating that they contribute additional independent information rather than duplicating existing variables. Overall, the correlation analysis did not reveal any excessively strong correlations that would suggest problematic multicollinearity among the engineered features. As a result, the derived variables were retained for further modeling.



Conclusion

This project explored the feasibility of developing a data-driven recommendation system for Magic: The Gathering Commander decks. The primary focus of this phase was acquiring a comprehensive dataset and conducting exploratory data analysis to better understand the structure and properties of the available card data.

Using the MTGJSON AllPrintings dataset, we constructed a filtered dataset of Commander-legal cards and analyzed key attributes such as mana value, power, toughness, rarity, colors, and card types. The analysis confirmed that the dataset contains a wide range of card characteristics that are useful for modeling deck composition and potential improvements. Summary statistics and distribution analyses provided insight into typical mana curves, the prevalence of different card types, and the balance of color identities within the dataset.

These findings demonstrate that the dataset contains sufficient information to support the development of a recommendation system. The exploratory analysis also helped identify several useful features that can be engineered to represent deck structure and strategic patterns. Overall, the results of this phase establish a strong foundation for building a similarity-based recommendation pipeline in the next stage of the project.

Next Steps

While the current phase of the project focused on data acquisition and exploratory analysis, the next stage will focus on developing the recommendation logic for the Commander deck recommender system. The exploratory analysis helped identify key card attributes such as mana value, color identity, card types, and rarity, which can be used to construct meaningful deck-level features. These features will be used to represent decks numerically so that similarities between decks and cards can be measured.

The next step will involve designing features that capture structural properties of Commander decks. Examples include the distribution of mana values, the balance of card types (creatures, instants, sorceries, etc.), the number of colors used in the deck, and indicators for specific strategic characteristics such as creature-heavy or spell-focused builds. These engineered features will allow decks to be compared using similarity measures such as cosine similarity or distance-based metrics.

Once deck representations are established, a recommendation framework can be implemented. The system will analyze a user-submitted decklist and compare it to existing card patterns in the dataset to identify potential improvements. Recommendations may include suggesting alternative cards with similar effects but stronger efficiency, identifying gaps in the deck's mana curve or card type balance, or recommending commanders that better align with the deck's strategy.

Finally, the project can be extended with a simple user interface that allows players to submit a decklist and view recommendations interactively. Additional improvements could include incorporating community-driven signals such as card popularity or synergy data from external sources like EDHREC, as well as evaluating the recommendations through feedback from Commander players during pilot testing.

Appendix A: GitHub Repository

The full code used for this project is available in the GitHub repository below:

GitHub Repository:
https://github.com/bnolan223/mtg-commander-recommender

The repository contains all scripts and notebooks used to perform the analysis presented in this report, including:

- Data preprocessing and cleaning
- Exploratory data analysis (EDA)
- Feature engineering
- Implementation of the analytical methods used in the study

Instructions for running the code and reproducing the analysis are provided in the repository's **README** file.

Appendix B: Dataset Description

Dataset Source

The dataset used in this project is derived from the **Magic: The Gathering AllPrintings dataset**, a comprehensive collection of card information across all official Magic: The Gathering sets. The data was filtered to include Commander-legal cards only, which are relevant to the objective of building a recommender system for Commander deck construction.

The original dataset was processed and cleaned using Python to remove duplicate printings and retain a single unique entry for each card.

Dataset Structure

After preprocessing and deduplication, the final dataset contains:

- 30,465 unique cards
- 11 variables

Each observation represents a unique Magic: The Gathering card, and the columns describe card attributes such as mana cost, card type, color identity, rarity, and combat statistics.

The dataset includes a mix of numerical, categorical, multivalued categorical, boolean, and text variables.

Variable Types

The dataset includes the following categories of variables:

- **Identifier variables**
  - `name`
  - `uuid`
- **Numerical variables**
  - `manaValue`
  - `power`
  - `toughness`
- **Categorical variables**
  - `rarity`

- **Multivalued categorical variables**
  - `colors`
  - `colorIdentity`
  - `types`
- **Boolean variable**
  - `isLegendary`
- **Text variable**
  - `text`

Dataset Characteristics

The dataset reflects the structural design of Magic: The Gathering cards:

- Creature cards represent the largest card type, accounting for approximately 53.6% of the dataset.
- Mono-colored cards dominate the dataset, representing approximately 72.5% of all cards, followed by multi-colored cards and colorless cards.
- Commons, Uncommons, and Rares make up the majority of cards, while Mythic and Special rarities are much less frequent.
- Approximately 11.34% of cards are legendary, which is particularly important in the Commander format since legendary creatures can serve as deck commanders.

Data Quality and Missing Values

Most variables in the dataset are complete. Missing values occur primarily in the power and toughness variables:

- `power` — 44.85% missing
- `toughness` — 44.69% missing

These missing values are structurally missing because non-creature cards do not have combat statistics.

The text variable contains approximately 1.15% missing values, which represent cards with no rules text rather than missing information.

Data Preparation

Several preprocessing steps were performed before analysis:

- Filtering the dataset to include Commander-legal cards
- Deduplicating card printings to retain one unique card per observation
- Converting numeric attributes such as power and toughness to numeric format
- Cleaning and preparing text data for later analysis
- Retaining only the variables relevant to exploratory analysis and feature engineering

The cleaned dataset was then used for exploratory data analysis and feature engineering to support development of a content-based Commander recommender system.

Appendix C: Data Dictionary

This section provides detailed descriptions of the variables included in the cleaned dataset and the derived features created during feature engineering.

Original Dataset Variables

Variable	Type	Description
`name`	String	Name of the Magic: The Gathering card.
`uuid`	String	Unique identifier assigned to each card entry in the dataset.
`manaValue`	Numeric	Converted mana cost representing the total resource cost required to cast the card.
`power`	Numeric	Offensive combat value of creature cards. Missing for non-creature cards.
`toughness`	Numeric	Defensive combat value of creature cards. Missing for non-creature cards.
`rarity`	Ordinal Categorical	Card rarity level with hierarchy: Common < Uncommon < Rare < Mythic < Special.
`colors`	Multivalued Categorical	List of colors associated with the card.
`colorIdentity`	Multivalued Categorical	Color identity used in Commander deck construction rules.
`types`	Multivalued Categorical	Card type categories such as Creature, Instant, Sorcery, Artifact, Enchantment, or Land.
`isLegendary`	Boolean	Indicator variable identifying whether a card has the Legendary supertype.
`text`	Text	Rules text describing the card's abilities and effects.

Engineered Variables

Several additional variables were created to support statistical analysis and machine learning modeling.

Variable	Type	Description
`num_colors`	Numeric	Number of colors associated with a card, derived from `colorIdentity`.
`text_length`	Numeric	Length of the rules text used as a proxy for mechanical complexity.
`color_W`	Binary	Indicator variable representing whether the card includes the White color identity.

color_U	Binary	Indicator variable representing whether the card includes the Blue color identity.
color_B	Binary	Indicator variable representing whether the card includes the Black color identity.
color_R	Binary	Indicator variable representing whether the card includes the Red color identity.
color_G	Binary	Indicator variable representing whether the card includes the Green color identity.
power_num	Numeric	Numeric representation of the creature power attribute.
toughness_num	Numeric	Numeric representation of the creature toughness attribute.
rarity_ord	Numeric (Ordinal)	Ordinal encoding of rarity categories to reflect rarity hierarchy.

Variable Categories for Modeling

The variables used in correlation analysis and modeling include:

Numerical Variables

- manaValue
- num_colors
- text_length
- power_num
- toughness_num
- rarity_ord

Binary Indicator Variables

- color_W
- color_U
- color_B
- color_R
- color_G

These engineered features allow card attributes such as color identity, rarity, mechanical complexity, and combat strength to be incorporated into similarity calculations for the recommender system.

Table of Contributions

The table below identifies contributors to various sections of this document.

	Section	Writing	Editing
1	**Analysis the basic metrics of variables**	**Bennett Nolan**	**Bennett Nolan**
2	**Non-graphical and graphical univariate analysis**	**Darryl Fields**	**Darryl Fields**
3	**Missing value and outlier analysis**	**Nate Moore**	**Nate Moore**
4	**Feature engineering and analysis**	**Dominic Durning**	**Dominic Durning**
5	**Appendix**	**Darryl Fields**	**Darryl Fields**